



If-else, if-elif-else, and match-cse in Python



With Coding

Outcomes of this Lecture

Use of else clause

Importance of if-elif-else

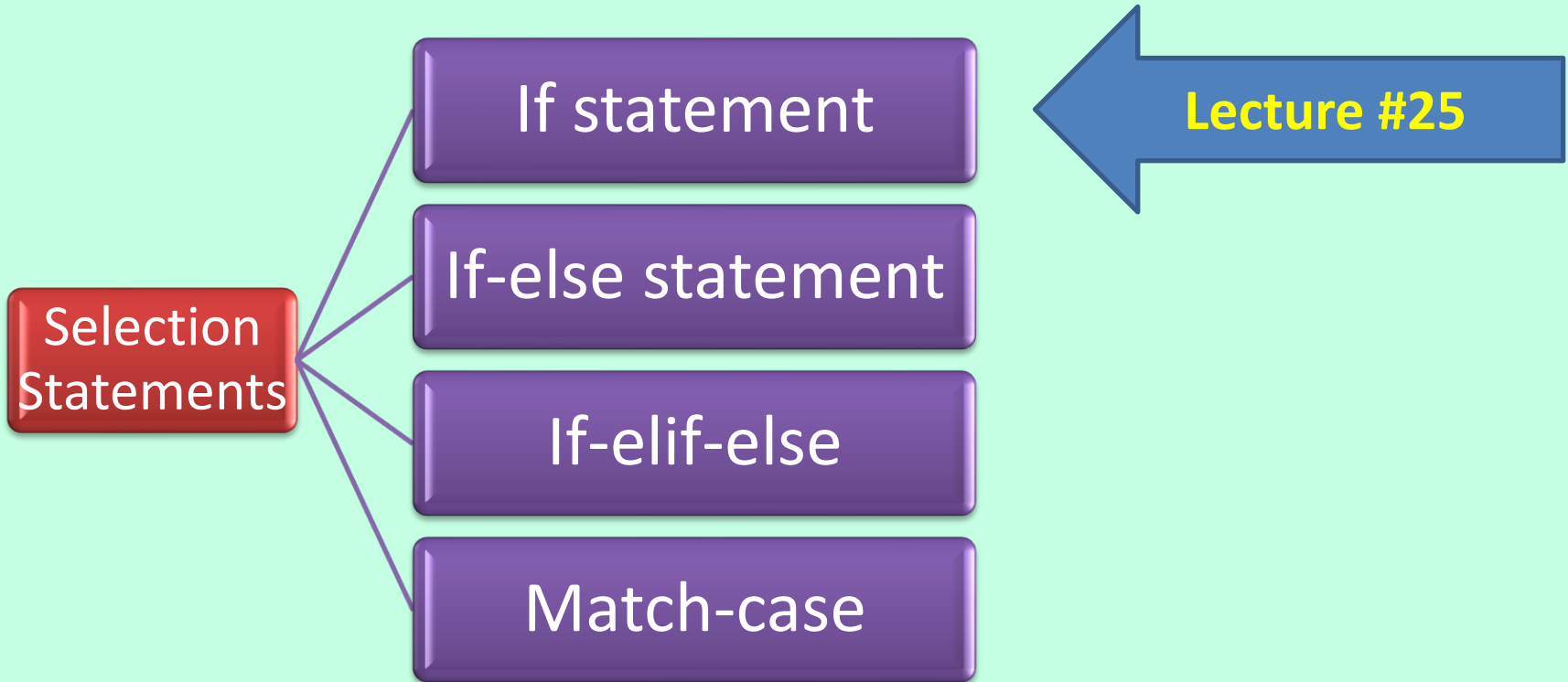
Execution procedure of if-elif-else

Implementation of if-else and if-elif

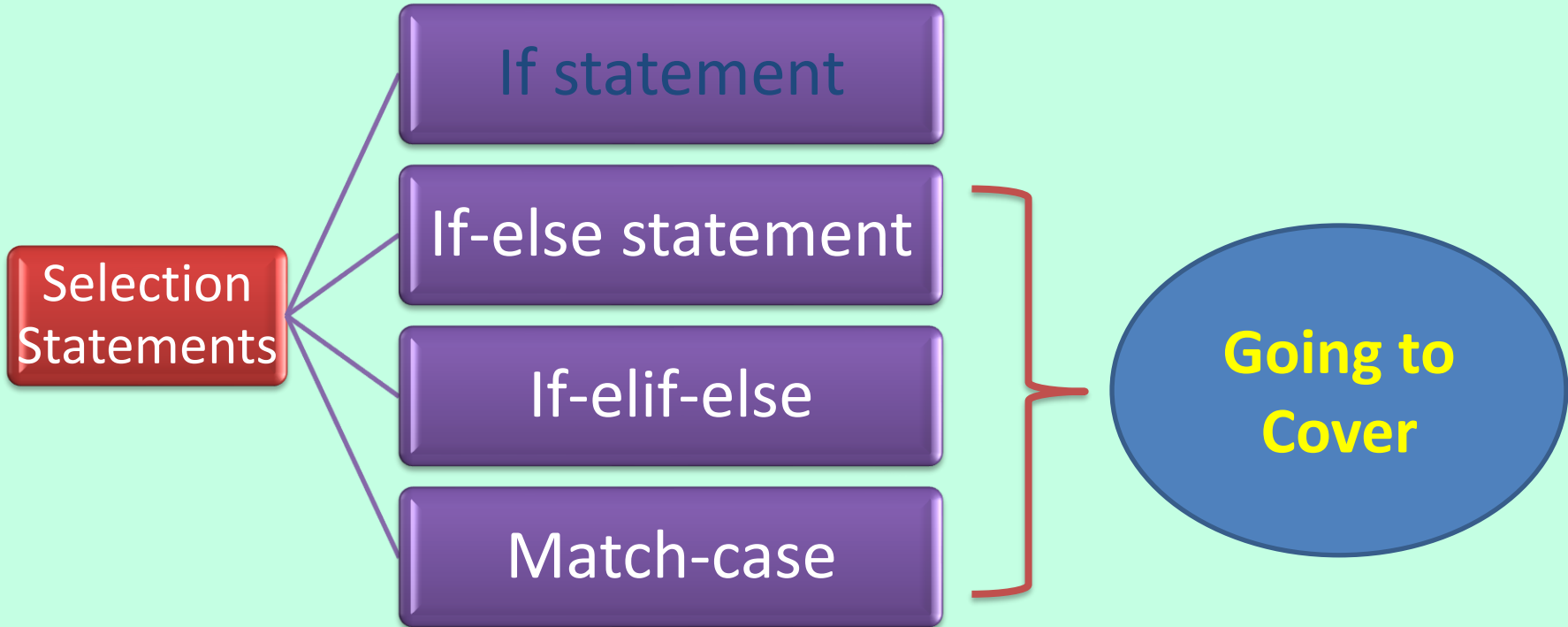
Importance match-case



Types of Control Statements



Types of Control Statements contd..





If-else

- This offers an alternative action.
- If the condition is true, the if block runs. If it's false, the else block runs.
- There is always an outcome
- If-else is **two-way decisions structures**.



If-else

- **Syntax:**

if condition:
statement(s)

else:
statement(s)

Real-life usage:

A security system.

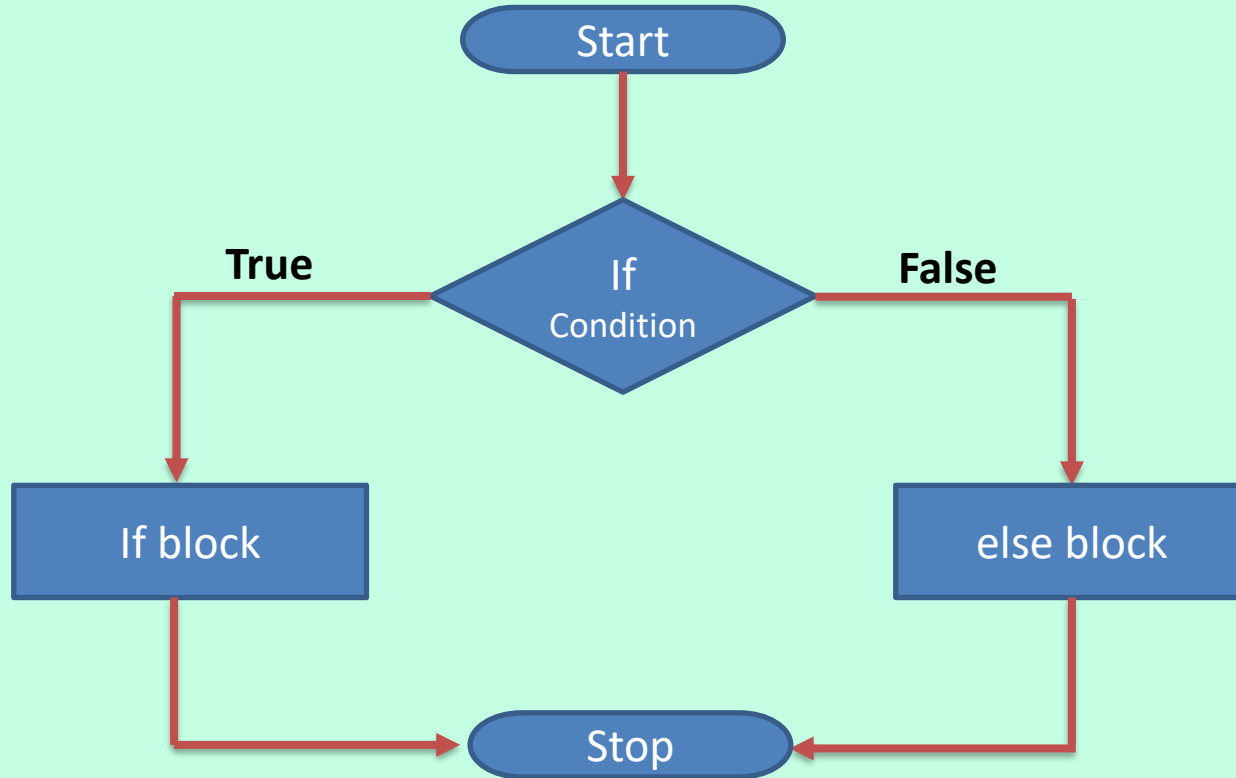
If the entered PIN is correct,

1. unlock the door.

2. *Else*, sound an alarm.



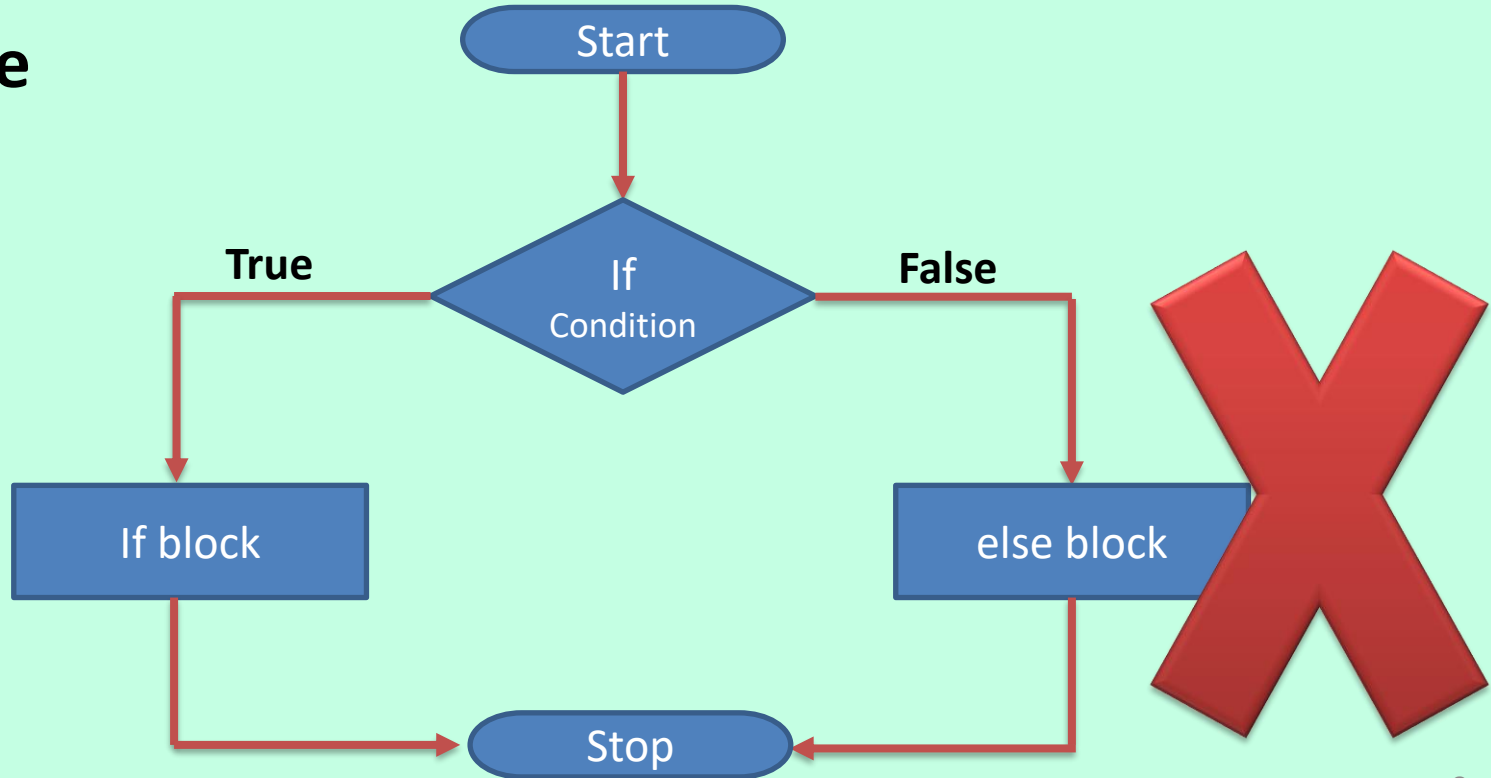
If-else Contd..





If-else Contd..

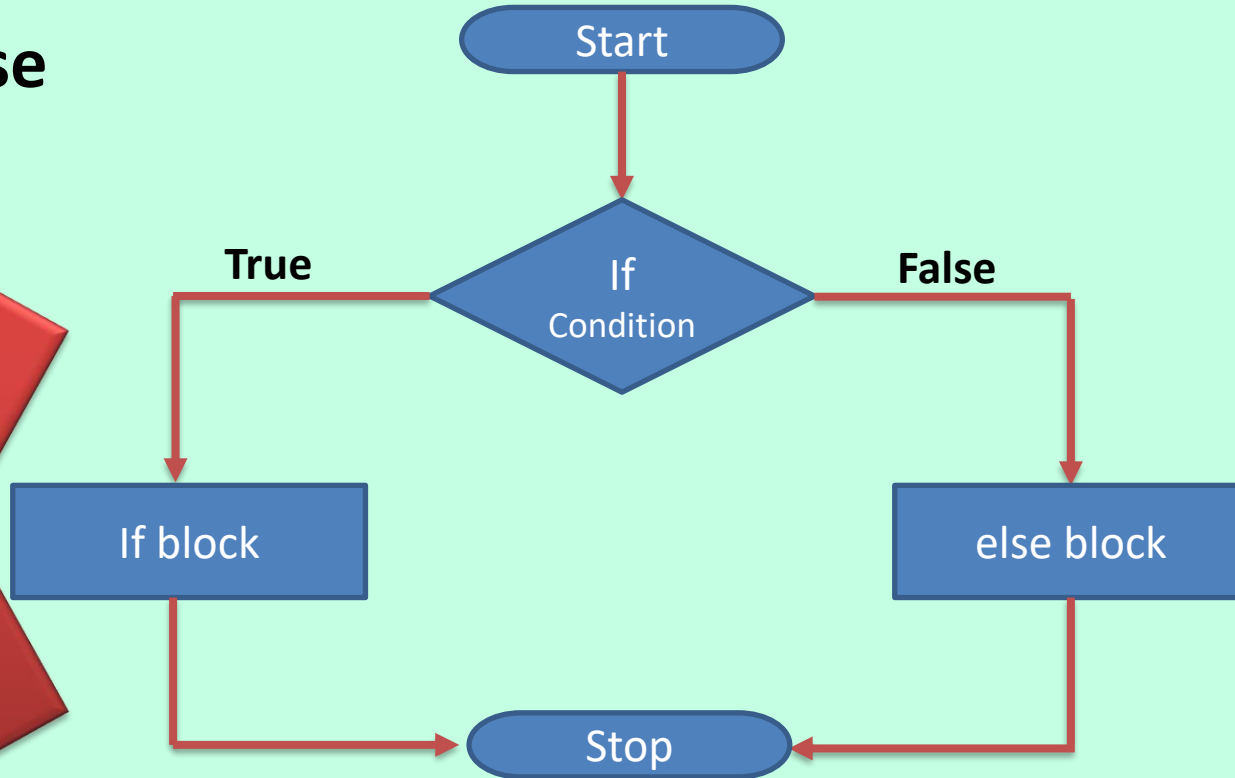
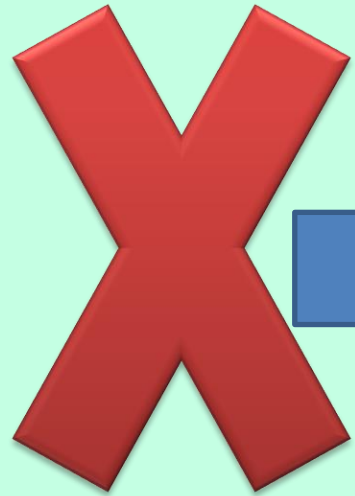
Case: True





If-else Contd..

Case: False





If-elif-else

- **elif** is a short form for **else if**.
- It is **multi-way decision** structure.
- In branching statements, when you want set **several alternatives**. For this, we use one or more elif clauses.
- For this we need to check multiple conditions in a sequential order.



Syntax: If-elif-else

if condition1:

Statement(s) to execute if condition1 is true

elif condition2:

Statement(s) to execute if condition1 is false and condition2 is true

elif condition3:

Statement(s) to execute if condition1 & 2 are false, and condition3 is true

else:

Statement(s) to execute if all conditions are false

Note: else part is optional



If-elif-else Contd..

- The first condition that is true gets its block executed, and the rest are ignored.
- If all the conditions are false, else will be executed
- If an else clause is not included and all the given conditions are false, in this case **none of the blocks** will be executed.



If-elif-else Contd..

Real-life usage: Grading a test.

1. If score > 90 , grade is A.
2. If score > 80 , grade is B.
3. If score > 70 , grade is C.
4. Else, the grade is D.



Match-case

- The **match-case** statement, introduced in Python 3.10, is a powerful structural pattern matching tool.
- While it may look like a traditional switch-case from other languages, it is significantly more capable because it can match complex patterns, types, and even specific data structures.



Syntax

match variable:

case pattern1:

action

case pattern2:

action

case _:

default action



Match-case

How it Works

- A match statement takes an expression and compares its value to successive patterns given as one or more case blocks.

Key Components:

- **match expression:** The variable or value you are inspecting.
- **case <pattern>:** The specific condition to check against.
- **_ (Underscore):** The "wildcard" pattern. It acts as a **catch-all** (default case) if no other patterns match.



Match-case

How it Works Contd..

Case Patterns can be:

- **Literal values** ("pizza", 42)
- **Data structures** (like tuples, lists)
- **Classes** (matching object attributes)
- **Guards** (conditions with if)

Example with guard:

```
match age:  
    case x if x < 18:  
        print("Minor")  
    case x if x >= 18:  
        print("Adult")
```



Applications

- match-case shines when you have **multiple discrete options** to handle, especially in apps like:
 - ❖ Command-line tools
 - ❖ Chatbots
 - ❖ API response handling
 - ❖ Menu-driven programs



Why use match-case over if-elif?

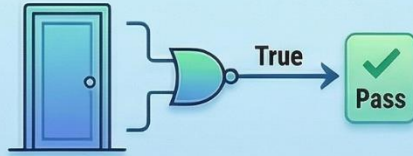
Feature	if-elif-else	match-case
Primary Use	Complex boolean logic (e.g., $x > 5$ and $y < 10$)	Structural patterns and specific value matching
Readability	Can get "wordy" with many conditions	Extremely clean and "declarative"
Catch-all	Uses the final else:	Uses case <code>_</code> :

PYTHON SELECTION STATEMENTS

(Decision Making)

BASIC SELECTION

if Statement



```
if condition:  
    execute()
```

if-else Statement

- Python 3

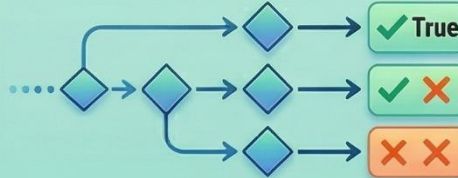
```
if condition:  
    do_this();  
else:  
    do_that()
```

MULTIPLE CONDITIONS

if-elif-else Ladder

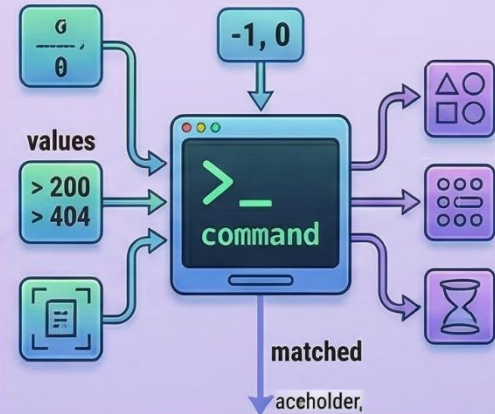


```
if cond1:  
    do_a();  
elif cond2:  
    do_b();  
else:  
    do_default()
```



STRUCTURAL MATCHING

match-case (Python 3.10+)



```
match status:  
    case 200: handle_success();  
    case 404: handle_not_found();  
    case _: handle_error()
```

Check your knowledge

When will the else block be executed?

At most, how many code blocks will be executed in if-elif?

Does else block mandatory in if-elif?

How to work with match-case

